Code No: 117DX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December-2023/January-2024
INFORMATION RETRIEVAL SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1.a)	Define Smoothing.	[2]
b)	Explain the following terms with examples i) tf_{ij} ii) df_j iii) idf_j	[3]
c)	What do you mean by Thesauri?	[2]
d)	What is the use of N-gram in retrieval?	[3]
e)	Define parsing.	[2]
f)	Discuss semantic networks.	[3]
g)	How to improve IRS efficiency?	[2]
h)	Define Inverted Index Modifications.	[3]
i)	How to Preprocess semi-structured test?	[2]
j)	Explain about the term Index.	[3]

PART-B

(50 Marks)

2. Find the similarity coefficient and similarity measures of vector space model for the following query
QUERY: "march health awareness"
D1: "the Health Observances for March"
D2: "the Health oriented Calendar"
D3: "the Awareness News for March Awareness" [10]

OR

3. Explain Non-Binary Independence model with an example that satisfies your statement. [10]

4. Explain the partial Query Expansion with different sort orders. [10]

OR

5. Explain about Hierarchical Agglomerative Clustering and its methods. [10]

6. What is Distance measure? Explain any two types in it. [10]

OR

7. Explain the concept of choosing translations with its approaches. [10]

8. Explain Fixed Length Index Compression with a suitable example. [10]

OR

9. Explain about the Throughput-optimized Compression and Block-addressing compressed indexes. [10]

10. What is Boolean Retrieval? Explain with example. [10]

OR

11. Explain in detail about Web Search. [10]

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